

MATTHEW TOPHAM

AI Systems Architect | Full-Stack Software Engineer

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SUMMARY

AI systems architect and full-stack developer building interactive storytelling, multimodal generation, and local-first agent systems. Active Marinara Engine developer with 15+ merged pull requests across React, TypeScript, Node.js, prompt orchestration, media pipelines, and persistent game state. Computer science student at CU Boulder experienced in turning product ideas into tested end-to-end features.

EXPERIENCE

Pasta Devs | Marinara Engine

2026 - Present

AI Systems Architect & Open-Source Developer | Remote

- Authored 15+ merged pull requests for a local-first AI chat, roleplay, and game engine, shipping across the React/TypeScript client, Node.js services, shared contracts, persistence, prompts, documentation, and tests.
- Built the end-to-end Game Mode storyboard and scene-video workflow: narrative-to-keyframe planning, image/video generation, galleries, automatic rendering, responsive playback controls, and archived-session replay.
- Designed prompt and provider architecture for still versus animation presets, keyframe count and duration controls, per-character NovelAI captions, campaign art styles, and editable preflight prompt review.
- Improved reliability through regression suites, lint/type/build checks, bundle-budget validation, manual browser QA, defensive timeouts, request queues, and user-facing technical documentation.

SELECTED PROJECTS

School of the Ancients | Creator & Full-Stack Developer

- Built a voice-driven Socratic learning platform where users converse with AI historical mentors across web and VR experiences.
- Implemented React/Vite interfaces, Node.js and Supabase services, authentication, RLS-backed Postgres, lesson and quest creation, CI/CD deployment, real-time voice AI, and Unity XR prototypes.

AI Radar | Creator

- Created a Python and GitHub Actions pipeline that ingests RSS feeds, filters and summarizes AI research/news with LLMs, and publishes a concise daily brief.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Bash

Frontend: React, Vite, Tailwind CSS, responsive web applications, PWAs

Backend & Data: Node.js, Express, Supabase, Postgres, REST APIs, WebSockets, SignalR, OAuth

AI & Systems: LLM integration, prompt architecture, agent workflows, multimodal image/video pipelines, local AI deployment

DevOps & Tools: Git, GitHub Actions, Docker, Linux, pnpm, Vercel, Unity

EDUCATION & CERTIFICATION

University of Colorado Boulder | B.A. Computer Science

Expected May 2027

Front Range Community College | A.S. Computer Science

2025

NVIDIA-Certified Associate | AI Infrastructure & Operations

Aug 2025